

A study of an Airbnb Shadow Inventory: NYC Housing Affordability & Tourism Economy

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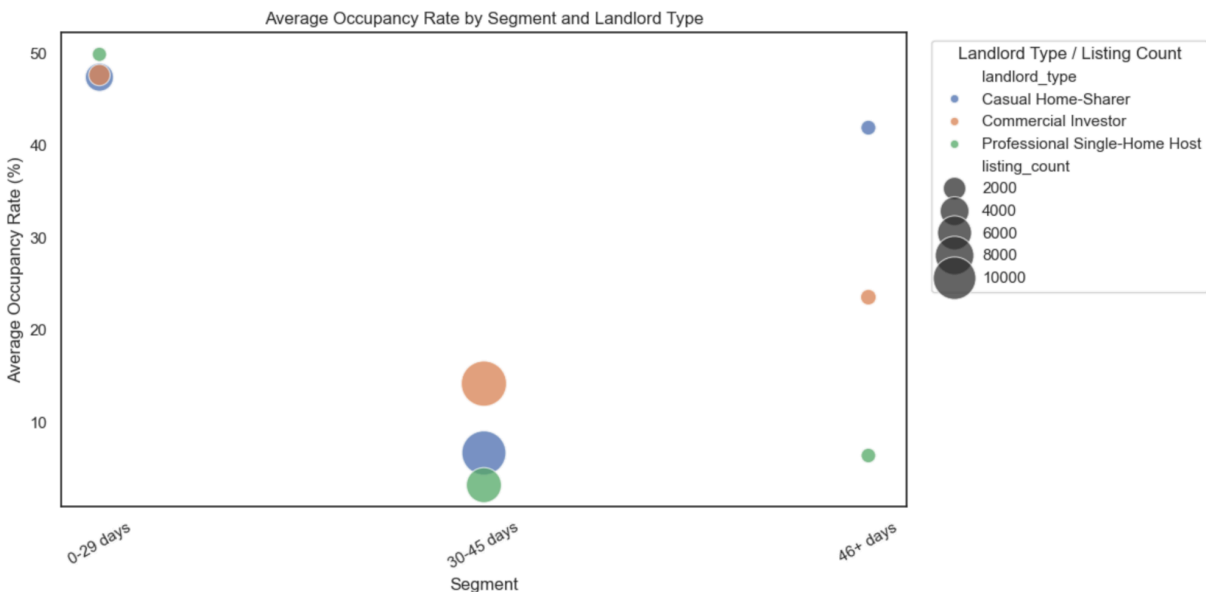
Executive Summary

Context & Purpose

I've always been curious about the relationship between short-term rentals and residential housing affordability in New York City. The Inside Airbnb data scraped on April 14, 2026 gives me the perfect opportunity to examine the operational behavior of platform hosts to determine if commercial real estate entities are artificially constricting long-term rental supply, thereby shifting local residential dynamics.

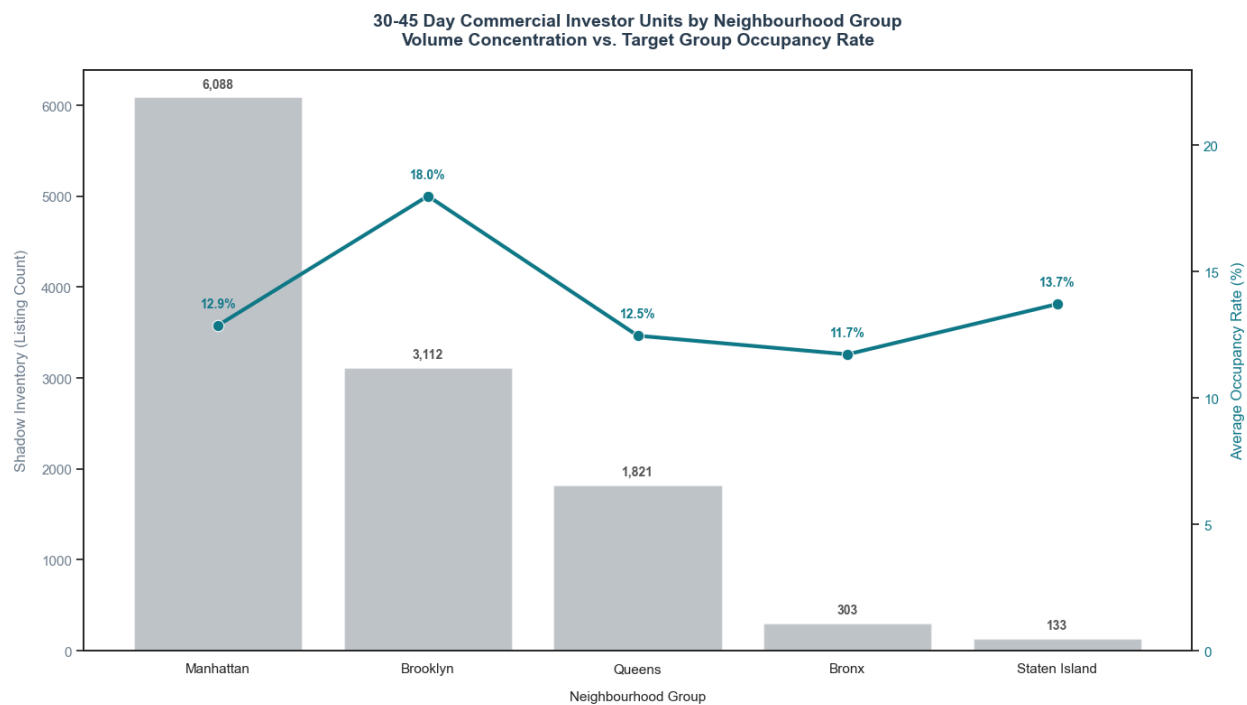
Core Findings #1

The data validates a highly inefficient market distortion. While traditional short-term rentals (under 29 days) maintain high utilization rates (~50% occupancy), they represent a minority segment of active listings (shown in figure below). Instead, platform volume is heavily saturated within the 30–45 day medium-term segment. Driven by NYC's strict short-term rental laws (Local Law 18), commercial operators have pivoted to this mid-term window to bypass regulatory registration.



These 30–45 day commercial listings suffer from low utilization, idling at average occupancy rates under 15%. This structural insulation creates a pool of "shadow inventory"—entire residential homes kept vacant and withheld from standard one-year leases, without generating the consistent tourist turnover or local neighborhood spending required to justify the inventory loss.

Core Findings #2



Geographically, this displacement pressure is asymmetric. Manhattan serves as the primary epicenter, hoarding over 6,000 underutilized shadow units. Brooklyn represents a substantial secondary cluster with over 3,100 units; however, its higher utilization rate (18.0% occupancy) signals a healthier integration compared to the severe stagnation seen across the outer boroughs.

Macroeconomic Implications

With thousands of airbnb residential units being underutilized in medium-term holdings, commercial real estate investors compress NYC’s vacancy rates and exacerbate rent inflation. The resulting premium on housing prices crowds out resident discretionary spending in local businesses, increases local corporate recruitment friction, and acts as a direct structural tax on the broader New York City workforce.